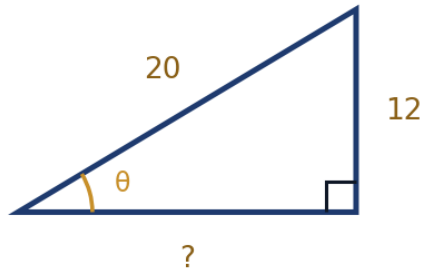


Trigonometry & Additional Topics



Right-triangle trigonometry

- 1 In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin \theta$, $\cos \theta$, and $\tan \theta$.



- 2 In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.
- 3 A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

The Pythagorean identity

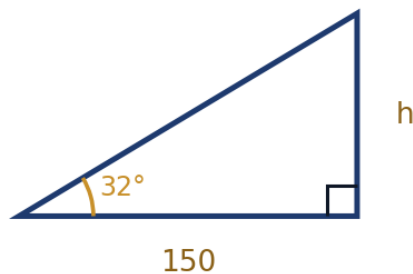
- 4 Given $\sin \theta = \frac{5}{13}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- 5 Given $\cos \theta = \frac{8}{17}$ with θ acute, find $\sin \theta$.

Exact trigonometric values

- 6 State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.
- 7 State the exact value of $\tan(60^\circ)$ and $\sin(90^\circ)$.

Solving for missing sides and angles

- 8 A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32° , as shown below. Find the height of the tower, to the nearest foot.



- 9 From the top of a 60-foot cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

Complex numbers: arithmetic

- 10 Add $(4 + 3i) + (2 - 7i)$.
- 11 Multiply $(3 + 2i)(1 - 4i)$.
- 12 Simplify i^7 , using $i^2 = -1$.

Complex numbers: modulus and conjugates

- 13 Find the modulus $|6 - 8i|$.
- 14 Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

Vectors: magnitude and operations

- 15 Vector $v = (3, 4)$. Find its magnitude.
- 16 Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.
- 17 Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

Radian measure

18 Convert 150° to radians, in terms of π .

19 Convert $\frac{7\pi}{4}$ radians to degrees.

A well-designed word problem: measuring the width of a river

20 This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C . (a) At point C , the angle between the bank and the line of sight to the tree is 35° . Use the tangent ratio to find AB , the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24° . Use this second measurement to find a second estimate of AB , to the nearest meter, and state which measurement gives the larger estimate.